UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the May/June 2010 question paper

for the guidance of teachers

5070 CHEMISTRY

5070/42 Paper 4 (Alternative to Practical), maximum raw mark 60

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1 **(a)** (gas) syringe (1)

(b) lime water turns milky / cloudy / white / chalky ppt. (1)

(c) 72 (1) cm^3

(d) 0.003 (1)

(e) **(i)** 0.003 (1)

(ii) 100 (1)

(iii) 0.3 (1) g

(f) 84 (1) 0.3 / 84 × 24000 = 85.7(86) (1) cm^3 [9]

2 **(a)** **(i)** shiny, silver or grey (1) (solid)

(ii) blue solution / liquid (1)

(b) **(i)** beaker gets warm or wtte (1)

(ii) copper (1) (accept Cu but not Cu(II))

(c) zinc dissolves / disappears; blue colour fades / disappears; fizzes / bubbles / effervescence / gas evolved. Any 2 (2)

(d) **(i)** $Zn + CuSO_4 = ZnSO_4 + Cu$ (1)

(ii) redox, displacement, reduction and / or oxidation (1) [8]

3 **(c)** [1]

4 **(b)** [1]

5 **(c)** [1]

6 **(b)** [1]

7 **(c)** [1]

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8 **(a)** pipette (1)

(b) yellow to pink, orange or red (1)

(c)

22.8	39.7	31.3
0	17.5	8.9
22.8	22.2	22.4

[Mark rows or columns to the benefit of the candidate. One mark for each correct row or column] (3)
Mean value 22.3 (1) cm^3

(d) 0.001 (1)

(e) 0.002 (1)

(f) 2 (1)

(g) 2 (1)

(h) **(i)** H_2SO_4 or sulfuric acid (1) (no H_2A)

(ii) $H_2SO_4 + 2NaOH = Na_2SO_4 + 2H_2O$ (1)
(ecf on **(g)** and **(h)(i)**) [12]

9 **(a)** transition metal / element / d-block but <u>not</u> V is a transition metal (1)

(b) **(i)** green ppt. (1)

(ii) insoluble in excess (1)

(iii) gas turns litmus blue (1) ammonia (1)

(c) aq. barium chloride / nitrate + hydrochloric / nitric acid (2)
white ppt. (1)

(d) aq. silver nitrate / nitric acid (2) white ppt. (1)
In parts **(c)** and **(d)** no acid or 'acidified' can score 2/3
White ppt. on own or no $BaCl_2$ or $AgNO_3$ no marks
No $Pb(NO_3)$ test.
Use of $BaSO_4$, $AgCl$, H_2SO_4 or HCl in test white ppt. mark only
Conclusion: NH_4Cl / $(NH_4)_2SO_4$ (1) $FeCl_2$ / $FeSO_4$ (1) [13]

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10 **(a)** 32, 55, 69, 80 all correct (2), one error (1)

(b) all points stated in **(a)** plotted correctly (1)
straight line (use of ruler) and smooth curve (1)
appropriate extrapolations at lower ends (1) and upper ends (1)

(c) **(i)** 0.35 g (1)

(ii) 2.6 g (1)

(d) 75°C (1)

(e) 35 g / 100 g of water (1)
For **(c)**, **(d)** and **(e)** results must be seen on graph.

(f) sodium chloride – colourless solution or no solid present (1)
potassium chlorate(V) – solid and liquid present (1) or some solid dissolved (not 'all solid undissolved') or wtte in both cases.
Mark individually.

(g) increase in temperature gives a large increase in solubility of potassium chlorate(V) but not much effect on solubility of sodium chloride (1) wtte. (Comparison required)

In parts **(c)**, **(d)** and **(e)** read candidate's graph in awarding marks.
Read graphs to + / – half small square. [13]

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